1 One-Way Functions

Recall our formal definition of a one-way function:

Definition 1.1. A function $f : \{0,1\}^* \to \{0,1\}^*$ is *one-way* if it satisfies the following conditions.

- *Easy to compute.* There is an efficient algorithm computing f. Formally, there exists a (uniform, deterministic) poly-time algorithm F such that $\mathsf{F}(x) = f(x)$ for all $x \in \{0,1\}^*$.
- *Hard to invert.* An efficient algorithm inverts f on a random input with only negligible probability. Formally, for any PPT algorithm $\mathcal{I}$, the *advantage* of $\mathcal{I}$

$$\mathbf{Adv}_f(\mathcal{I}) := \Pr_{x \leftarrow \{0,1\}^n}\left[\mathcal{I}(1^n, f(x)) \in f^{-1}(f(x))\right] = \mathrm{negl}(n)$$

  is a negligible function (in the security parameter n).

We also allowed a syntactic relaxation where the domain and range (and the input distribution over the domain) may be arbitrary, as long as the input distribution can be sampled efficiently (given the security parameter n).

1.1 Candidates

At the end of last lecture we defined two functions, with the idea of investigating whether they could be one-way functions:

1. Subset-sum: define $f_{\text{ss}} : (\mathbb{Z}_N)^n \times \{0,1\}^n \to (\mathbb{Z}_N)^n \times \mathbb{Z}_N$, where $N = 2^n$, as

$$f_{\text{ss}}(a_1, \ldots, a_n, b_1, \ldots, b_n) = (a_1, \ldots, a_n, S = \sum_{i\,:\,b_i=1} a_i \bmod N).$$

   Notice that because the a_is are part of the output, inverting f means finding a subset of $\{1, \ldots, n\}$ (corresponding to those b_is equaling 1) that induces the given sum S modulo N.

2. Multiplication: define $f_{\text{mult}} : \mathbb{N}^2 \to \mathbb{N}$ as[1]

$$f_{\text{mult}}(x, y) = \begin{cases} 1 & \text{if } x = 1 \vee y = 1 \\ x \cdot y & \text{otherwise.} \end{cases}$$

First consider f_{ss}; clearly, it can be computed in polynomial time. It syntactically matches subset-sum, the famous NP-complete problem, and is conjectured to be one-way.

Question 1. Recall that subset-sum is an NP-complete problem, so, assuming $\mathsf{P} \neq \mathsf{NP}$, it has no polynomial-time algorithm. Does this prove that f_{ss} is a one-way function?

Now consider f_{mult}, which also can be computed in polynomial time. Syntactically, inverting f_{mult} means factoring xy into non-trivial integer factors.

[1] For security parameter n, we use the domain $[1, 2^n] \times [1, 2^n]$. The cases $x = 1$ and $y = 1$ are special to rule out the trivial preimages $(1, xy)$ and $(xy, 1)$ for the output $xy \neq 1$.

Question 2. Factoring integers is conjectured to be a hard problem. Does this assumption imply that f_{mult} is a one-way function?

The answer is no: f_{mult} is *not* a one-way function as we have defined it, and this is due to the difference between factoring in the worst case versus the average case. While *some* integers seem hard to factor, a large fraction of integers are very easy to factor. Concretely, consider the following inverter $\mathcal{I}(z)$: if z is even, then output $(2, z/2)$, otherwise fail. Notice that $\mathcal{I}$ outputs a valid preimage of z exactly when $z = xy$ is even, which occurs with non-negligible probability $3/4$ when x, y are chosen independently and uniformly from $[1, 2^n]$ (because xy is even exactly when at least one of x, y is even).

There are two common ways to deal with this difference between the worst case and average case. The first is to "tweak" the function—actually, its input distribution—to focus on the instances that we believe to be hard. For the factoring problem, it is believed (but not proved!) that the hardest numbers to factor are those that are the product of exactly two primes of about the same size. More specifically, define

$$\Pi_n = \{p \ : \ p \in [1, 2^n] \text{ is prime}\}$$

to be the set of "n-bit prime numbers." Then we may make the following plausible conjecture:

Conjecture 1.2 (Factoring Assumption). For every PPT algorithm $\mathcal{A}$,

$$\Pr_{p,q \leftarrow \Pi_n}[\mathcal{A}(p \cdot q) = (p, q)] = \text{negl}(n).$$

Let's now change the input distribution of f_{mult} so that x, y are chosen uniformly and independently from Π_n. Now, the one-wayness property for f_{mult} is syntactically identical to Conjecture 1.2, so for that property we have a "proof by assumption." The only remaining question is the distribution over inputs: can we efficiently sample a random n-bit prime, i.e., sample uniformly from Π_n? Algorithm 1 gives a potential algorithm, GenPrime, to do so.

Algorithm 1 Algorithm $\mathsf{GenPrime}(1^n)$ for sampling a uniformly random n-bit prime.

1: Choose a fresh uniformly random integer $p \in [1, 2^n]$.
2: If p is prime, output p. Otherwise start over.

Clearly, if GenPrime terminates, it outputs an element of Π_n. Moreover, it outputs a *uniformly random* element of Π_n, because every $\bar{p} \in \Pi_n$ is equally likely to be chosen in the first step. So the only remaining question is the running time: can we test primality efficiently? And can we guarantee that the algorithm terminates quickly, i.e., does not start over too many times?

The first question has many answers. The definitive answer was provided in 2002 by Agrawal, Kayal, and Saxena, who gave a *deterministic* primality test that runs in time polynomial in the bit length of the number (i.e., $\text{poly}(n)$ for a number in $[1, 2^n]$). The running time, while polynomial, is quite large, so the AKS test is not used in practice. Instead, *randomized* tests such as Miller-Rabin are used to quickly test primality; however, such tests have a *very small* probability of returning the wrong answer ("prime" when the number is composite, or vice versa, or both). This introduces a tiny bias into the output distribution of GenPrime, which we can fold into a negligible error term that we won't need to worry about.[2] (Later on we will rigorously define what "negligible error term" means. We may study primality tests in more detail later.)

[2] Another alternative is to use *deterministic* primality tests which only have proofs of correctness under unproved (but widely believed) number-theoretic assumptions. These are less common in practice because they aren't so fast.

The second question, about the running time of GenPrime, is also very interesting. The probability that a particular iteration terminates is exactly $|\Pi_n|/2^n$ (assuming a perfect primality test). How does $|\Pi_n|$ compare to 2^n, i.e., how "dense" are the n-bit primes? It is easy to show that there are an infinite number of primes, but this is not strong enough for our purposes—primes could be so vanishingly rare that GenPrime effectively runs forever. Fortunately, it turns out that the primes are quite dense. For an integer $N > 1$, let $\pi(N)$ denote the number of primes not exceeding N. We have the following:

Lemma 1.3 (Chebyshev). *For every integer $N > 1$, $\pi(N) > \frac{N}{2\log_2 N}$.*

(In fact, this is a weaker form of the "prime number theorem," which says that $\pi(N)$ asymptotically approaches $\frac{N}{\ln N}$. We will not give a proof of Lemma 1.3 now, but there is an elementary one in the Pass-shelat notes.) Using Lemma 1.3, we have

$$\frac{|\Pi_n|}{2^n} = \frac{\pi(2^n)}{2^n} > \frac{2^n}{2^n \cdot 2\log_2(2^n)} = \frac{1}{2n}.$$

Finally, since all the iterations of GenPrime are independent, the probability that it fails to terminate after (say) $2n^2$ iterations is

$$(1 - \tfrac{1}{2n})^{2n \cdot n} \leq (1/2)^n,$$

which is exponentially small.[3]

1.2 Weak OWFs and Hardness Amplification

The second way of dealing with the above difficulty of "somewhere hard" functions is more general, but also most costly and technically more complex. Suppose we are not confident where the "hard" inputs to a OWF are likely to be, but we do believe that they aren't too rare. We would then have a *δ-weak* one-way function, where the "hard to invert" property from Definition 1.1 is relaxed to the following:

- For a specified $\delta = \delta(n) = 1/\operatorname{poly}(n)$, every PPT algorithm $\mathcal{I}$ satisfies, for all sufficiently large allowed n,
$$\mathbf{Adv}_f(\mathcal{I}) := \Pr_{x \leftarrow \{0,1\}^n}\left[\mathcal{I}(f(x)) \in f^{-1}(f(x))\right] \leq 1 - \delta(n).$$

The only change is that we have replaced the $\operatorname{negl}(n)$ success probability by $1 - \delta$. (To contrast with the "δ-weak" nomenclature, we sometimes say that the original condition defines a *strong* OWF.) This new definition allows for f to be easily invertible for a large fraction of inputs (e.g., all the even numbers, all those divisible by three, perfect squares, and the like), but every efficient inverter still fails with probability at least δ.

Suppose that f is some arbitrary δ-weak OWF. Can we always get a (strong) OWF from it? The answer is yes; this kind of result lies within the area called *hardness amplification*. For a function f, define its m-wise *direct product* as

$$f^{(m)}(x_1, \ldots, x_m) = (f(x_1), \ldots, f(x_m)),$$

where we parse the input so that $|x_1| = \cdots = |x_m| = n$. We allow $m = m(n)$ to depend on n and write $f' = f^{(m)}$ for short when m is clear from context or its precise value is immaterial. The total input length $N = n \cdot m(n)$ is the security parameter for f'. So, the allowed security parameters are precisely these lengths, as n ranges over $\mathbb{N}$ (or, if applicable, the allowed security parameters of f). To ensure that the total length determines how to parse the input efficiently, we require $m(n)$ to be positive integer-valued, nondecreasing,

[3] To be completely formal, in order to say that GenPrime is polynomial time, we should modify it so that if it has not terminated after $2n^2$ iterations, it just outputs some fixed value (say, 1). This introduces only an exponentially small bias in its output distribution.

and polynomial-time computable from 1^n. Then $nm(n)$ is strictly increasing, and the unique component length n can be recovered in polynomial time in N by trying lengths from 1 through N. Thus f' is efficiently computable by parsing the input and evaluating f on each component.[4]

Theorem 1.4. *If f is a δ-weak one-way function, then for any such $\mathrm{poly}(n)$-bounded $m(n) \geq 2n/\delta(n)$, the direct product $f' = f^{(m)}$ is a (strong) one-way function for its allowed security parameters.*

Such a choice of m exists because $1/\delta(n) = \mathrm{poly}(n)$, thus we can take $m(n) = Cn^k$ for sufficiently large positive integers C, k.

Question 3. Does inverting $f'(x_1, \ldots, x_m)$ require inverting $f(x_i)$ for all $i = 1, \ldots, m$? Since a PPT algorithm can have advantage at most $1 - \delta$ in inverting a single $f(x_i)$, does this immediately imply that a PPT algorithm has advantage at most $(1-\delta)^m \leq (1-\delta)^{2n/\delta} \leq (1/e)^{2n} = \mathrm{negl}(n)$ in inverting f'?

Proof idea. The intuition is that inverting f' requires inverting f on m independently chosen values. If each component is inverted independently with probability at most $1 - \delta$, then the probability of inverting all of them should be at most

$$(1-\delta)^m \leq (1-\delta)^{2n/\delta} \leq 1/e^{2n} = \mathrm{negl}(n).$$

Rigorously proving that this intuition is correct, however, turns out to be quite subtle. The reason is that an adversary $\mathcal{I}'$ attacking f' does not have to treat each of the inversion subtasks $y_1 = f(x_1), \ldots, y_m = f(x_m)$ independently. For example, it might look at all of the y_i at once, and either invert *all* of them (with some non-negligible probability overall), or none of them. So the individual events (that $\mathcal{I}'$ succeeds in the sub-task of inverting $y_i = f(x_i)$) can be highly correlated, and we cannot simply say that the overall success probability is bounded by $(1-\delta)^m$.

We will show how to use any PPT inverter $\mathcal{I}'$ for f' to construct a PPT inverter $\mathcal{I}$ for f, which "amplifies" the success probability $\alpha = \alpha(n)$ of $\mathcal{I}'$. By the weak one-wayness of f, however, no PPT inverter can succeed with probability greater than $1 - \delta$ for infinitely many allowed parameters. Together, these facts will force α to be negligible, as desired.

The basic idea of the reduction is this: $\mathcal{I}$ is given $y = f(x)$ and wants to use $\mathcal{I}'$ to invert y. An obvious first attempt is to "plug in" the value y as part of the output of f', e.g., to run $\mathcal{I}'$ on

$$y' = (y_1 = y = f(x), y_2 = f(x_2), \ldots, y_m = f(x_m)) = f'(x_1 = x, x_2, \ldots, x_m)$$

for some random $x_2, \ldots, x_m$ chosen by $\mathcal{I}$. When $\mathcal{I}'$ outputs a candidate preimage $x' = (x'_1, \ldots, x'_m)$, our $\mathcal{I}$ just outputs x'_1 as its candidate preimage of $y = f(x)$. Notice that if $\mathcal{I}'$ happens to invert y' successfully, then in particular $x'_1 \in f^{-1}(y)$, as required.

Unfortunately, this simple idea does not quite work. This gives $\mathcal{I}$ success probability at least α, but we want to amplify its success probability toward 1. We might hope to fix this problem by running $\mathcal{I}'$ a large number of times, using the same $y_1 = f(x)$ but *fresh* random $x_2, \ldots, x_m$ each time, expecting $\mathcal{I}'$ to successfully invert at least once. (Note that we can check whether $\mathcal{I}'$ succeeded in inverting f' on a particular tuple, because f' is efficiently computable.) However, repetition does not necessarily increase the success probability: for a fixed challenge y, the success probability of each attempt may be much smaller than α, or

[4] If the allowed security parameters of f are all of $\mathbb{N}$, we can extend f' to all input lengths: on an input of length $L \geq m(1)$, apply f' to the longest prefix of length $nm(n)$ and discard the remaining bits; for the finitely many shorter lengths, output the empty tuple. If f' is one-way for its allowed input lengths and $m(n)$ is polynomially bounded, a generic input-truncation argument for OWFs shows that the extension is also one-way, since $nm(n) \leq L < (n+1)m(n+1) = \mathrm{poly}(n)$.

even zero. Indeed, $\mathcal{I}'$ might behave in a "sneaky" way to make this very strategy fail. For example, $\mathcal{I}'$ might have a certain fixed α-fraction of "good" values of x_1 for which it always inverts (ignoring $x_2, \ldots, x_m$), and always fail for *all* other values of x_1. Notice that such $\mathcal{I}'$ has overall advantage α, as required. But no matter how many times we repeatedly run $\mathcal{I}'$ with $y_1 = y = f(x)$, we will eventually succeed only if the original x is "good," which happens with probability only α.

A better reduction (which will turn out to work) just applies the above "plug-and-repeat" strategy for *every* position $i = 1, \ldots, m$. As it turns out, unless α is already exponentially small, *some* position has a very large ($> 1 - \delta/2$) fraction of "good" x_is, for a suitable definition of good. Conditioned on the x from our given instance $y = f(x)$ being good, repeated calls to $\mathcal{I}'$ with fresh randomness and fresh other components drive the probability of success toward 1. We will quantify how the number of repetitions depends on α, and use the weak one-wayness of f to bound α. We now give the formal proof.

Proof of Theorem 1.4. Fix any PPT inverter $\mathcal{I}'$, and write $\alpha(n)$ for its advantage on f' at input length $N = nm(n)$. Our goal is to show that $\alpha(n) = \mathrm{negl}(n)$. We first analyze a fixed n, suppressing the dependence of α, m, δ on n.

To identify inputs on which repetition is effective, define the "good" set for position i by

$$G_i = \left\{ x_i : \Pr_{x_j \text{ for } j \neq i} \left[\mathcal{I}'(f'(x_1, \ldots, x_m)) \text{ succeeds} \right] \geq \frac{\alpha}{2m} \right\},$$

where the other components are independent and uniform in $\{0,1\}^n$, and the probability also includes the random coins of $\mathcal{I}'$.

Claim 1.5. *If $\alpha > 1/2^{n-1}$, then some G_i has density greater than $1 - \delta/2$ in $\{0,1\}^n$.*

Proof. By a union bound and independence of the uniformly random components,

$$\begin{aligned} \alpha &\leq \Pr[\text{every } x_i \in G_i] + \sum_{i=1}^{m} \Pr[\mathcal{I}' \text{ succeeds and } x_i \notin G_i] \\ &\leq \prod_{i=1}^{m} \frac{|G_i|}{2^n} + m \cdot \frac{\alpha}{2m}. \end{aligned}$$

Hence the product of the densities is at least $\alpha/2$, so at least one density is at least $(\alpha/2)^{1/m}$. This exceeds $1 - \delta/2$, because

$$(1 - \delta/2)^m \leq e^{-m\delta/2} \leq e^{-n} \leq 2^{-n} < \alpha/2. \qquad \square$$

Our inverter $\mathcal{I}(1^n, y = f(x))$ tries every position i, making some $t \geq 1$ (to be determined below) independent attempts for each position. In each attempt, it places y in position i, samples the other components independently, and calls $\mathcal{I}'$ with fresh random coins and security parameter 1^N. If the returned tuple is a valid preimage, $\mathcal{I}$ outputs its ith component; if all attempts fail, $\mathcal{I}$ reports failure. Each attempt takes polynomial time in n, including computing 1^N and running $\mathcal{I}'$, since $N = \mathrm{poly}(n)$. There are at most mt attempts, so $\mathcal{I}$ is PPT whenever t is polynomially bounded and efficiently computable. (The algorithm $\mathcal{I}$ does not need to know α or the sets G_i.)

Suppose $\alpha > 1/2^{n-1}$, and take a position i supplied by Claim 1.5. With probability greater than $1 - \delta/2$, the input x lies in G_i. For each such x, all t attempts at this position fail with probability at most $(1 - \alpha/(2m))^t \leq e^{-t\alpha/(2m)}$. Therefore,

$$\mathbf{Adv}_f(\mathcal{I})(n) \geq 1 - \frac{\delta}{2} - e^{-t\alpha/(2m)}.$$

If $\alpha(n)$ were non-negligible, then $\alpha(n) \geq n^{-c}$ for some constant $c > 0$ and infinitely many allowed n. Choose a polynomial repetition count $t(n) > 2m(n)n^c \ln(2/\delta(n))$, which is possible because $m(n)$ and $1/\delta(n)$ are polynomially bounded. For sufficiently large such n, we have $\alpha(n) > 1/2^{n-1}$ and $e^{-t\alpha/(2m)} < \delta/2$. The above bound then makes the advantage of $\mathcal{I}$ greater than $1 - \delta$ for infinitely many allowed parameters, contradicting the weak one-wayness of f. Therefore, $\alpha(n) = \mathrm{negl}(n)$, as desired. Finally, since $N = \mathrm{poly}(n)$ and $N \to \infty$ with n, Lecture 2's change-of-parameter property makes the advantage of $\mathcal{I}'$ negligible in N, proving that f' is one-way. $\square$

Answers

Question 1. Recall that subset-sum is an NP-complete problem, so, assuming $\mathsf{P} \neq \mathsf{NP}$, it has no polynomial-time algorithm. Does this prove that f_{ss} is a one-way function?

Answer. No. The fact that subset-sum is NP-complete means that, assuming $\mathsf{P} \neq \mathsf{NP}$, no polynomial-time algorithm solves subset-sum in the *worst case*, i.e., on *any* given instance. However, in the one-way function inversion game, the input to f_{ss} is chosen uniformly *at random*, and is not necessarily a "hardest" instance of subset-sum. In other words, an inverter for f_{ss} only needs to solve subset-sum *on the average* and with good enough probability, not necessarily in the worst case and with probability one. While we *conjecture* that almost all instances of subset-sum are hard, and hence f_{ss} is one-way, as far as we know it could be the case that a large fraction of subset-sum instances are easy to solve, yet a rare few are hard. This would allow a polynomial-time inverter to attain non-negligible advantage against f_{ss} without solving the NP-complete subset-sum problem in the worst case.

Question 2. Factoring integers is conjectured to be a hard problem. Does this assumption imply that f_{mult} is a one-way function?

Answer. No! In fact, f_{mult} is certainly *not* a one-way function as we have defined it, for the reasons given following the text of this question.

Question 3. Does inverting $f'(x_1, \ldots, x_m)$ require inverting $f(x_i)$ for all $i = 1, \ldots, m$? Since a PPT algorithm can have advantage at most $1 - \delta$ in inverting a single $f(x_i)$, does this immediately imply that a PPT algorithm has advantage at most $(1-\delta)^m \leq (1-\delta)^{2n/\delta} \leq (1/e)^{2n} = \text{negl}(n)$ in inverting f'?

Answer. The answer to the first question is yes: any preimage of $f'(x_1, \ldots, x_m) = (f(x_1), \ldots, f(x_m))$ is a tuple $(x_1^*, \ldots, x_m^*)$ of equal-length strings x_i^* where $f(x_i^*) = f(x_i)$, i.e., each x_i^* is a preimage of $f(x_i)$. (This argument assumes that the output tuple of f' is encoded so that it is uniquely parseable into its m components.)

The answer to the second question is no: that reasoning is not valid, because while the components $y_i = f(x_i)$ are independent, an adversary attacking them *need not treat each of the inversion subtasks independently*; it might correlate its successes and failures. See the text following the original question for further elaboration, and the proof of Theorem 1.4 for a rigorous argument.